Hospital Operations & Billing Intelligence

Analysis Report

Power BI Intern Final Project

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Tools: Power BI, Power Query, DAX, Excel

Reporting period: 8 May 2019 to 7 May 2024

Date: 23 October 2026

GitHub repository: Hospital-Operations-Billing-PowerBI

1. Project Overview

Project: Hospital Operations & Billing Intelligence (Power BI dashboard).

Objective: To explore hospital admission data, clean and model it, and build an interactive Power BI dashboard that shows admissions, billing and length of stay, so that operations and billing teams can see what is happening and where they can improve.

Business problem: Hospital managers need one clear view of how many patients are admitted, how much is billed, how long patients stay, and how test results and admission types are spread. Without it, it is hard to plan beds, check billing quality and decide where to look for improvement.

Target audience: Hospital operations managers, billing and finance teams, and senior decision-makers.

Dataset: Kaggle healthcare_dataset (CC0 licence), 55,500 raw rows, 50,000 rows after cleaning.

2. Dataset Summary

Item	Detail
Source	Kaggle healthcare_dataset (CC0 public domain)
Raw size	55,500 rows, 15 columns
Cleaned size	50,000 rows, 18 columns (Admissions_Clean)
Grain	One row = one patient admission
Time period	8 May 2019 to 7 May 2024 (2019 and 2024 are partial years)
Key fields	Admission Date, Discharge Date, Medical Condition, Insurance Provider, Admission Type, Billing Amount, Test Results, Age, Medication
Characteristics	Values are spread almost evenly across insurers, conditions and age groups, so the data appears synthetic. Currency is not stated.

3. Dataset Assessment & Cleaning

Every column was assessed in *Data/assessment/Column_Assessment.xlsx*. The raw data was kept unchanged in *Data/raw*. All cleaning was done in Power Query, and the steps are saved in *Scripts/PowerQuery_Steps.m*.

Finding	Action taken	Reason
534 exact duplicate rows	Removed (first copy kept)	Same admission recorded twice
About 4,966 near-duplicates that differ only in Age	Removed (first copy kept)	Same record with a changed age value
Patient Name column	Removed	Privacy; not needed for analysis
96 rows with negative billing (total -47,650.40)	Kept and flagged in Billing Status	Possible refunds or adjustments; deleting would hide real events
No Admission ID	Added Admission ID	Gives each admission a unique key
Stay length not available	Added Length of Stay (Day) = Discharge Date minus Admission Date	Needed for operations analysis
Age as a number only	Added Age Group (18-34, 35-49, 50-64, 65+)	Easier to compare groups
Currency not stated	Documented, not assumed	Avoids a wrong unit

Result: 55,500 raw rows minus about 5,500 duplicates gives 50,000 clean rows.

4. Data Model Overview

The model is a **star schema**: one fact table in the centre, filtered by seven small dimension tables. This keeps the model simple, fast and easy to explain, and it lets every slicer filter the measures correctly.

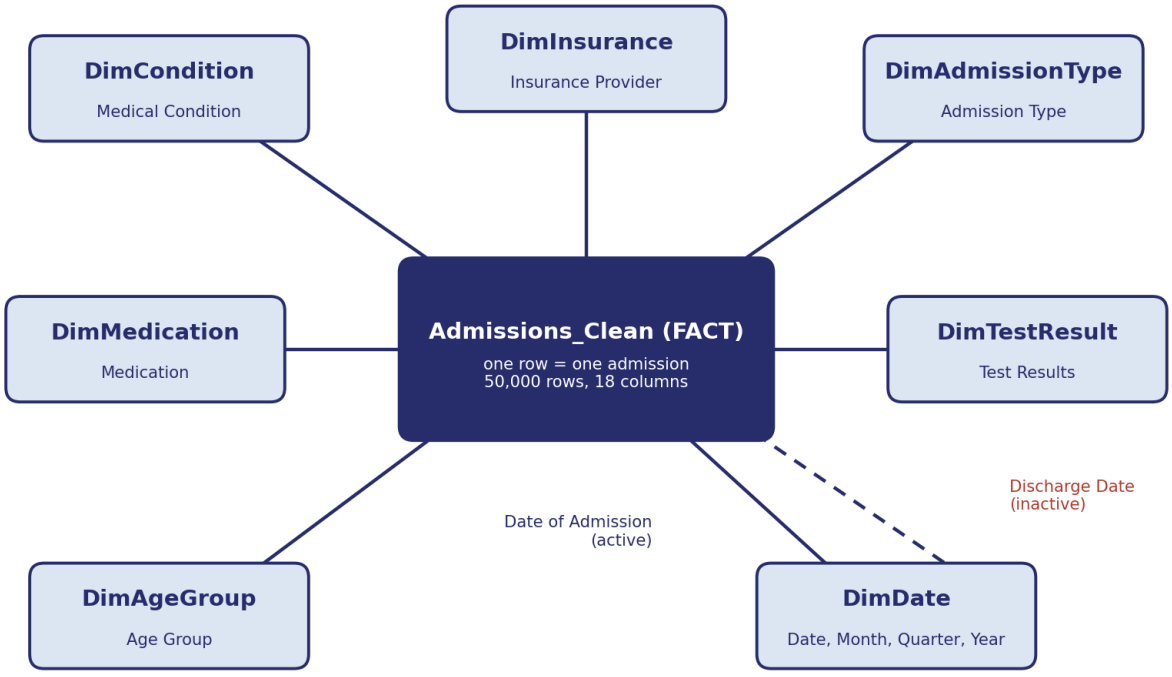


Figure 1: Star schema of the Hospital Operations & Billing model.

Tables and purpose

Table	Type	Purpose
Admissions_Clean	Fact	One row per admission: dates, billing amount, length of stay, billing status and keys
DimCondition	Dimension	Medical Condition list for slicers and charts
DimInsurance	Dimension	Insurance Provider list
DimAdmissionType	Dimension	Elective, Urgent, Emergency
DimMedication	Dimension	Medication list
DimTestResult	Dimension	Normal, Abnormal, Inconclusive
DimAgeGroup	Dimension	Age bands (18-34, 35-49, 50-64, 65+)
DimDate	Dimension	Date table with Month, Month Number, MonthStart, Quarter and Year
_Measures	Measure table	Holds all DAX measures in one place

Relationships

- Each dimension connects to Admissions_Clean with a **one-to-many** relationship (one dimension row, many admissions), with **single-direction** filtering from the dimension to the fact table.
- **DimDate to Date of Admission** is the **active** relationship, so all time charts use admission date.
- **DimDate to Discharge Date** is **inactive**. A model can have only one active path between two tables, so discharge date is kept for use with explicit DAX when needed.

Key measures

Measure	Business definition
Total Admissions	Count of admissions (one row = one admission)

Measure	Business definition
Total Billing	Sum of Billing Amount (currency not stated)
Avg Billing per Admission	Total Billing divided by Total Admissions
Avg Length of Stay	Average of Discharge Date minus Admission Date, in days
Emergency Admission %	Emergency admissions divided by total admissions
Abnormal Test Result %	Abnormal results divided by total admissions
Negative Billing Count	Number of admissions with a negative billing amount
Billing Equivalent / Bed Days Freed	What-if: effect of shortening every stay by the chosen number of days. Assumes billing is proportional to stay days
Dynamic Title, Selected Condition	Text measures that show the current filter selection

Measures are used for values that must change with filters. Calculated columns (Length of Stay, Age Group) are used for row-level values that do not change with filters.

Assumptions and modelling limits

- Billing is treated as the billed amount, not cost or profit.
- Patient names were removed, so repeat patients cannot be tracked.
- Because the data looks synthetic, relationships between fields (for example condition and billing) are not realistic.

5. Analysis & Insights

1. **Overall volume:** The hospital handled **50,000 admissions** between 8 May 2019 and 7 May 2024, with total billing of **1,277.8M** and an average of **25,555.69** per admission.
2. **Billing is almost the same across insurers.** Average billing ranges only from 25,413 (UnitedHealthcare) to 25,651 (Blue Cross), a gap of about 0.9%. This is too small to call a real difference.
3. **Admission types are evenly split:** Elective 33.65%, Urgent 33.57%, Emergency 32.78% (about 16,400 emergency admissions).
4. **Length of stay is about 15.5 days in every group.** Age groups and medical conditions look nearly the same, and stays from 1 to 30 days are spread evenly.
5. **One in three tests is abnormal:** Abnormal is 33.5%, and Normal and Inconclusive are close to a third each.
6. **Shorter stays have the biggest scenario impact.** Reducing every stay by 1 day would free 50,000 bed days, with a billing equivalent of about 82.4M. This assumes billing is proportional to stay days, so it is a scenario, not a forecast.
7. **Billing data quality:** 96 admissions (about 0.19%) have negative billing, totalling -47,650.40. They look like refunds or adjustments and are very small compared with total billing.

Overall pattern: No insurer, condition or age group stands out. Differences are under 1%. This is itself a finding, and it is one reason the data is thought to be synthetic.

6. Business Implications

Insight	Why it matters to the hospital
Billing is the same across insurers	No insurer needs special pricing or contract action based on this data.
Stay length is about 15.5 days everywhere	Length of stay is a general process issue, not a problem of one condition or age group. Improving discharge flow can help all patients.
1-day shorter stay frees 50,000 bed days	Even a small reduction can release a large amount of bed capacity, which supports more admissions or lower waiting times.
Emergency is about one-third of admissions	Emergency capacity (beds, staff) must be kept ready all year, not only in peaks.
One-third of tests are abnormal	A large share of patients need follow-up, which affects staff time and bed use.
96 negative billing records	Small in value, but they must be explained, because unexplained refunds can point to billing errors.

7. Conclusions

- The dashboard gives one clear view of admissions, billing, length of stay and test results for 50,000 admissions.
- The data is very evenly spread, so differences between insurers, conditions and age groups are too small to act on.
- Length of stay is the clearest lever for operational improvement, based on the what-if scenario.
- All key numbers were checked against the cleaned source data, and no differences were found.
- Better data (stated currency, full years, cost data) is needed before making real hospital decisions.

8. Recommendations

#	Recommendation	Linked finding
1	Treat length of stay as the main lever. Study discharge delays first.	Insights 4 and 6
2	Review the 96 negative billing records with the billing team and confirm whether they are refunds, corrections or errors.	Insight 7
3	Plan emergency capacity at about one-third of admissions, since the share is steady.	Insight 3

#	Recommendation	Linked finding
4	Set up a clear follow-up process for abnormal test results.	Insight 5
5	Do not target actions by insurer, condition or age group yet. The data shows no clear difference.	Insights 2 and 4
6	Collect better data before real decisions: state the currency, include full years, and add real cost data.	Limitations

9. Dashboard Overview

Page	Purpose	Main content
Overview	Quick summary for everyone	6 KPI cards, monthly trend with metric switch, admission type mix, dynamic title, key takeaway
Billing	Billing and finance view	Total and average billing, billing by insurer and by condition, insurer x admission type matrix, negative billing table
Operations	Operations view	Length of stay, abnormal test and emergency %, stay by age group and condition, stay distribution, test result mix, what-if slider
Definitions (hidden)	Explains every KPI	Definitions, data period and limitations; opened with the Info button
Tooltip (hidden)	Hover details	Total admissions, total billing and average stay on hover
Condition Details (hidden)	Drill-through page	Admissions and billing for one condition, with a record table

Filters and interactions

- **Slicers:** Year, Medical Condition, Insurance Provider and Admission Type on all pages; Age Group on Operations only.
- **Metric switch** on the monthly trend (Total Admissions, Total Billing, Avg Length of Stay).
- **Navigation buttons** (Overview, Billing, Operations), **Reset**, **Info** and **Hide filters**.
- **Cross-filtering** between visuals, a **tooltip page**, and **drill-through** to Condition Details with a Back button.
- **What-if slider** (Stay Reduction) on the Operations page.

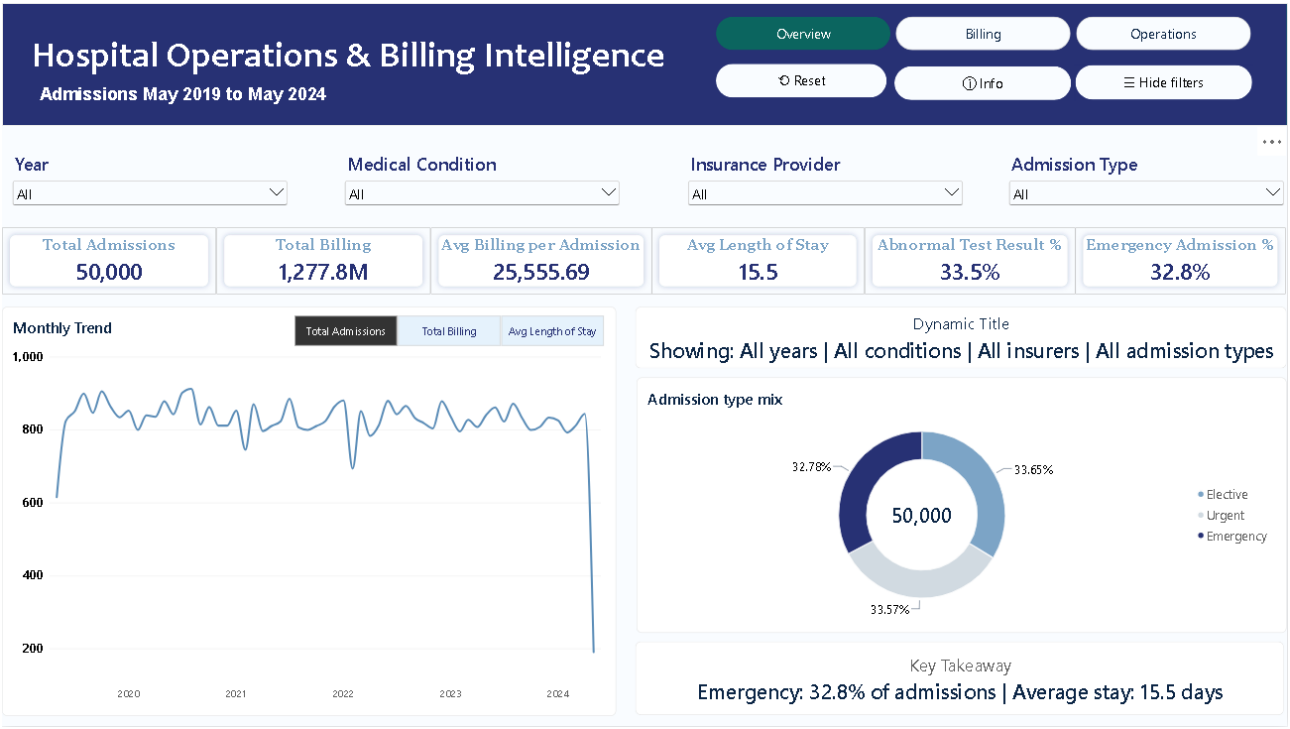


Figure 2: Overview page. The sharp drop at the end of the trend line is May 2024, which has only 7 days of data (the data ends on 7 May 2024). It is a partial month, not a real fall.

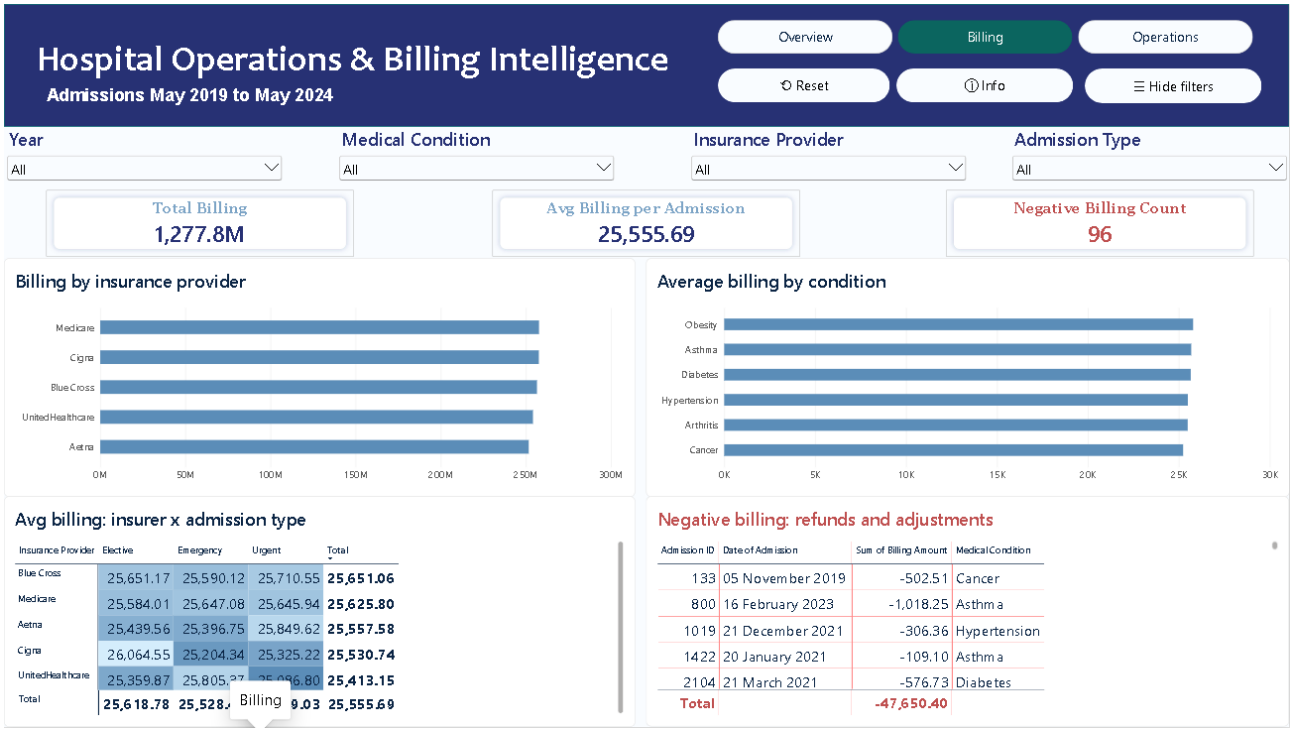


Figure 3: Billing page, showing billing by insurer and condition and the negative billing table.



Figure 4: Operations page, with length of stay views and the what-if slider.

10. Validation

Key values on the dashboard were compared with calculations done in Excel on the cleaned dataset (Data/assessment/Validation_Checks.xlsx). No discrepancies were found.

Check	Expected (cleaned data)	Dashboard	Result
Total admissions	50,000	50,000	Pass
Total billing	1,277,784,576.67	1,277.8M	Pass
Average billing	25,555.69	25,555.69	Pass
Average stay	15.5 days	15.5	Pass
Emergency admissions	32.8%	32.8%	Pass
Abnormal test results	33.5%	33.5%	Pass
Negative billing	96 rows (-47,650.40)	96 (-47,650.40)	Pass
Cleaning	55,500 - 534 exact - about 4,966 near-duplicates	50,000	Pass

Other tests: slicers (for example Year and Insurance Provider) gave the same totals as Excel; **Reset** returns every value to 50,000 admissions; and sums by insurer, condition and age group add up to the grand total.

11. Limitations & Assumptions

- **Currency is not stated**, so billing values have no unit.
- **2019 and 2024 are partial years** (8 May 2019 to 7 May 2024), so yearly totals cannot be compared directly. May 2019 and May 2024 are partial months.
- **The data appears synthetic**, because values are almost perfectly evenly spread. Findings may not match a real hospital.
- **About 5,500 duplicates were removed** (534 exact and about 4,966 near-duplicates that differed only in Age). The first copy was kept, so a few real repeat visits may have been lost.
- **96 negative billing rows were kept and flagged**, not deleted, so they slightly reduce total billing.
- **Billing is not profit or cost.**
- **The what-if assumes billing is proportional to stay days.** It is a scenario, not a forecast.
- **Patient names were removed** for privacy, so repeat patients cannot be tracked.